

# EBID

*An Online Auction & Marketplace Platform with Integrated Digital Wallet*

## Project Overview

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EBID is a modern e-commerce platform that combines standard online shopping with real-time competitive auctions. Instead of buying select high-demand or exclusive items at a fixed price, users bid against one another until a countdown timer expires, with the highest bid winning the item. To make the process fast and trustworthy, EBID includes a built-in Digital Wallet: users pre-deposit funds, and bid amounts are instantly frozen against their balance. This guarantees that every bid is backed by real money, removes the "unpaid item" problem common on platforms like eBay, and lets sellers receive payment automatically and immediately once an auction closes — combining the familiarity of eBay-style bidding with StockX-style focus on a fast, active marketplace for valuable goods.

## Team Members

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| Name   | Student ID |
|--------|------------|
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## Main Features

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- **Live Bidding:** Live Bidding – real-time view of the current highest bid, with instant updates as users place higher bids.
- **Auction Timers:** Auction Timers – every listed item has a strict countdown; the auction closes automatically when time expires.
- **Digital Wallet:** Digital Wallet – built-in banking system for deposits, balance tracking, automatic bid-fund freezing/release, and instant seller payouts.
- **Role System:** Role-Based Access – distinct experiences for Users (bidders), Sellers (item listers), and Admins (platform management).
- **Seller Tools:** Seller Tools – create auctions with photos, descriptions, starting price, and custom auction duration.

## Technologies to be Used

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- **Backend:** Spring Boot – REST APIs, business logic, and backend services.
- **Frontend:** React.js – dynamic, responsive single-page user interface.
- **Deployment:** Containerized services for consistent, portable deployment across environments.
- **Microservices Architecture:** Independent services (e.g., Auction, Wallet/Payment, User/Auth) for scalability and modularity, targeted for extra credit.